Industrial Safety Processes

Material Handling

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Injuries in Material Handling

- Account for about 20 to 45% of all occupational injuries
- Hands and fingers
- Damaged legs and feet
- Back problems
- Cuts and abrasions
- Muscle strains

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Hands

- Cuts and abrasions
- Caught between items carried/moved and fixed objects (walls, racking, piping)
- Crushing injuries due to items slipping and falling

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Preventing Hand Injuries

- Wearing proper gloves
- Ensuring proper grip on tools, objects being pushed/pulled, items carried
- Proper visibility to identify possible pinch points
- Planning the lift/move

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Feet

- Dropping items on feet
- Tripping, stubbing toes
- Kicking items not seen due to obstructed view
- Potentially being hit by other equipment – PITs, hand trucks

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Back Injuries

- Major issue in material handling
- Virtually impossible to accurately diagnose many “soft tissue” injuries
- Almost all diagnoses are based on patient reports of pain, not physically identifiable issues
- Can range from muscle fatigue and strains to pinched nerves to ruptured discs

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Back Belts

- Regularly available and a wide range of claims are associated with them
- Some controversy over their use, though they are becoming more common in material handling
- Very careful consideration should be made regarding the use of back belts in the workplace and whether an employer will permit them

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Back Belts

- Back belts are not recommended by Alberta Workplace Health and Safety – see Let`s Back Up a Bit – Some Truths about Back Belts
- Should not be used as a method to prevent back injuries
- Could make workers dependent on their use through weakening of muscles in back and abdomen
- May be used post-injury under the supervision of a doctor

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https://open.alberta.ca/dataset/30431817-7868-4b5b-b197-bfe6beac59fe/resource/8a5db5f5-0b01-4162-abb5-275f29630df9/download/2691963-2004-04-lets-back-up-a-bit-some-truths-about-back-belts.pdf

Let’s Back up a Bit

There is:

- little scientific evidence of increased lifting power,
- little scientific evidence of lower rates of injury in workers wearing back belts, and
- some evidence of potential harm from increased abdominal pressure and increased blood pressure.

Do not rely on back belts to prevent back injuries. Implement an injury prevention program based on workplace redesign and the tasks being performed.

Let’s Back Up a Bit – The Truth about Back Belts, AB WHS Bulletin PH003

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Guidelines for Lifting

- Many variables impact the “safety” of manually lifting objects
- See the NIOSH Lifting Equation for 7 dynamic variables
- Bottom line is that the activity must match the capabilities of the worker

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No “Rules” But…

- Eliminate manual lifting as much as possible
- Be in good physical shape
- Think before doing
- Get a good grip
- Lift with the legs
- Don’t twist or turn while lifting
- Keep the load close to your body

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Personnel Selection vs. Training

- Modelling and equations only work so well
- How and where lifting takes place is incredibly important
- Some people shouldn’t lift some things
- Training has had minimal effect on back injury rates over the last 40 years – yet training is considered a key factor

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From Week #1

"The improvement in system performance that can be realized from the redesign of equipment is usually greater than the gains that can be realized from the selection and training of personnel."

Alphonse Chapanis

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Team Lifting

- Get help with objects that are:
  - Too heavy for one person
  - Too awkward for one person
  - Too large for one person
- Plan the lift
- Try to keep it level
- Move in unison – especially with large groups

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Odd Sizes, Shapes & Materials

- Special glass handling racks and equipment
- Barrel movers
- Odd and unusual sizes and shapes

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Hand Tools

- Hooks
  - Sharp and possible puncture agents
  - Can bounce of hard objects and cause injuries
- Crowbars
  - May slip if not seated properly
  - Hands may slide if contaminated
- Rollers / Carts / Hand Trucks
  - Crushing hazard for fingers and toes, running up on heels

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Two-Wheeled Hand Trucks

- Don’t pile too high – maintain clear visibility
- Select unit capable of handling expected load
- Secure bulky items
- Chain/strap in place compressed gases
- Push, don’t pull behind – see what you are doing

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Four-Wheeled Trucks

- Same rules as two-wheeled
- Be more careful with loads
  - Tendency to overload
  - Tendency to pile too high – obstruct view
- Significant risk of catching hand against door jams, walls, corners

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Drums

- Can be heavy
  - typically 205 litres of chemical
  - Content density may vary
  - Partially filled drums may “slosh” and centre of gravity can change
- Odd shape to handle and store
- Often made from different materials
  - Poly drums, different steel drums

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Tanks

- Come in wide variety of sizes
- Can be more or less easy to move or store based on configuration and contents
- Type of contents will determine type and construction of tank
- Tank farms are special environments with special considerations not covered here

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Containers for Solids

- Bins, hoppers, silos
- Rolls
- Bags and bundles
- Racking vs. Stacking
- Piles
- Pallets
- Need to consider access, possible damage (to stock and by stock)

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Some Issues

- Dust
  - May contribute to hazardous atmospheres
  - Could interrupt with equipment operation
- Engulfment
  - In silos, large bins
  - From falling stacks of bags or bundles
- Mechanical means of moving can be limited for bags and bundles

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Pallets & Racking

- Maximum heights is an issue
  - Fragile items of particular concern
- Items falling from racks can be issue in warehouse environments
- Require equipment to move and lift
- Racking can suffer significant damage and fail

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Containers for Gases

- Compressed gas cylinders
- Some piping
- Cryogenic liquids
  - Super-cooled gases
  - Liquid nitrogen, liquid oxygen
- Primary issues relate to fire/explosion
- Issues with rupture due to mishandling

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Floors, Ramps, Aisles

- Should be sufficient clearance for expected types and size of loads
- Consider turning radius of any equipment used
- Keep clear of obstructions, slip/trip hazards, oils, etc.
- Changes in elevation should be noted

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Dock Boards

- Need to be strong enough to hold weight when moving material in and out of containers
- Should be able to handle 4x expected weights
- Need to take into consideration use of equipment when off-loading

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Strapping & Ties

- Can be dangerous when cut
- Injuries to hands are common
  - Possible lacerations to face, neck and arms
- Cut straps straight to avoid creating jagged and additional sharp edges
- Stay out of the “line of fire” for strapping under tension

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